Case 1: prior dominates (strong prior, little data) – NCS wildcat wells

θ = probability that a wildcat well on the Norwegian continental shelf makes a discovery.

- Prior (strong): 229 discoveries in 468 wells (2010–2024).

$$\theta \sim \text{Beta}(229,\ 239)$$

- Data (little): Barents Sea 2025, 5 discoveries in 6 wells (k = 5, n = 6).
- Posterior:

$$\theta \mid k \sim \text{Beta}(234,\ 240)$$

The 95% credible interval is [0.449, 0.539].

Note: the source gives the 2025 count as "wells drilled", while the prior counts wildcat wells only.

Case 2: likelihood dominates (weak prior, a lot of data) – Stephen Curry free throws

θ = probability that Stephen Curry scores from a free throw (1 point).

- Prior (weak): a flat prior, worth only 2 shots (1 made, 1 missed).

$$\theta \sim \text{Beta}(1,\ 1)$$

- Data (a lot): 4,259 made of 4,668 career free throws (k = 4,259, n = 4,668).
- Posterior:

$$\theta \mid k \sim \text{Beta}(4{,}260,\ 410)$$

The 95% credible interval is [0.904, 0.920].

Here k = 4,259 is the number of made free throws and n − k = 409 is the number of missed free throws.

The credible intervals differ because the two cases have different amounts of data and different prior distributions.

- Case 1 (NCS wildcat wells): The posterior 95% credible interval is relatively wide because only 6 new wells were observed. The strong prior helps constrain the estimate.
- Case 2 (Stephen Curry): The 95% credible interval is [0.904, 0.920], with a width of 0.016. It is narrow because 4,668 free-throw attempts provide substantial information.

In conclusion, Case 1 has greater uncertainty because the new dataset is small, while Case 2 has lower uncertainty because the large dataset produces a more precise estimate.